



Republic of the Philippines
TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES
Ayala Boulevard, Ermita, Manila, 1000
www.tup.edu.ph



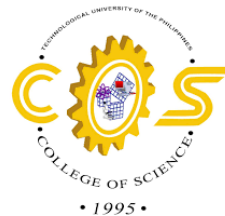
WEEK 4 : FIREBASE CRUD OPERATIONS LABORATORY

Kirby D. Ramilo

BSIT-2A

Prof. *Darwin Vargas*

March 21, 2025



Brief answers to the following:

1. What did you learn from this lab?

Through this lab, I learned how to implement CRUD (Create, Read, Update, Delete) operations using Firestore in a web application. I gained hands-on experience in integrating Firebase, handling Firestore collections and documents, and using JavaScript to interact with the database. Additionally, I explored real-time updates with Firestore's `onSnapshot()` function, which allows automatic data synchronization. Debugging various issues, such as connection errors and permission settings, also deepened my understanding of Firebase authentication and Firestore security rules.

2. What challenges did you encounter while connecting to Firestore?

One of the major challenges I faced was ensuring that the Firebase configuration in `script.js` matched my project settings correctly. Initially, Firestore operations failed due to incorrect API keys and database URLs. Another issue was setting up proper Firestore security rules, as restrictive settings prevented read and write operations. I also encountered CORS (Cross-Origin Resource Sharing) issues while running the project locally, requiring adjustments in browser settings. Additionally, handling real-time listeners properly was a challenge because multiple event listeners sometimes led to duplicate UI updates.

3. How does Firestore differ from traditional databases?

Firestore differs from traditional databases in several ways. Unlike relational databases that use structured tables and rows, Firestore follows a NoSQL approach, storing data in collections and documents. This makes it more flexible, as it does not require a predefined schema. Firestore also provides real-time synchronization, automatically updating the UI whenever data changes, whereas traditional databases require manual refreshing. Another key difference is scalability—Firestore is designed to handle distributed applications efficiently with built-in offline support. These features make Firestore ideal for modern web and mobile applications that require real-time data updates and dynamic structures.



Republic of the Philippines
TECHNOLOGICAL UNIVERSITY OF THE PHILIPPINES
Ayala Boulevard, Ermita, Manila, 1000
www.tup.edu.ph

